

Asymmetric Cascade Photocycloaddition-Acyloin Rearrangement Enabled by Cage-Confined Visible-Light Catalysis

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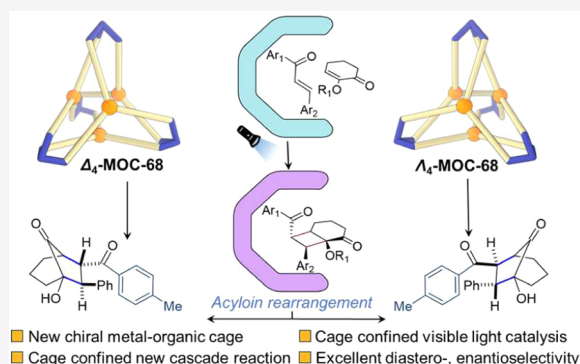
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ABSTRACT: Enzymatic catalysis in biological systems is characterized by the specific pocket confinement imposed by various protein matrixes, enabling the synthesis of a diverse array of functional biomolecules. Development of new catalysts that incorporate multiple catalytic centers within the enzyme-mimic confined spaces presents a meaningful yet challenging project for synthetic chemists. Here, we present our recent achievement in synthesizing a chiral photosensitive metal–organic cage (cPMOC), Δ_4 -/ Λ_4 -MOC-68- Ru_4 , which possesses multiple chiral pockets that can facilitate the visible-light-induced asymmetric cascade intermolecular [2 + 2] cycloaddition/acyloin rearrangement for the first time. The current photochemical transformation affords various bicyclo[3.2.1]octanes with opposite regioselectivity in contrast to well-established thermodynamically favored cascade [3 + 2] cycloaddition transformations with similar reaction counterparts. The distinctive pocket size of cage 2 inhibits the dimerization of α,β -unsaturated ketones and promotes the heterocycloaddition between the ketone and smaller cyclic 1,2-dione, underscoring the critical role of the microenvironmental shape and size for guest binding in determining the confined catalytic reactivity.



INTRODUCTION

The synthesis of biologically active molecules of complex structures in living organisms depends on a diverse range of enzymes.^{1–3} Various enzymes feature distinct substrate binding behaviors and catalytic pathways, facilitating the production of different biomolecules. For example, while members of the cytochrome P450 (CYP) family share similar “intrinsic reactivity”, differences in the size and shape of their binding pockets create varying equilibrium distributions of binding substrates, leading to varied product selectivity within CYPs.⁴ Metal–organic cages (MOCs) represent a type of biomimetic structures that possess guest-binding capabilities and potentially active sites, allowing them to replicate enzyme-mimicking catalytic functions.^{5–10} The shape and size of the binding cavities within MOCs significantly influence the chemo-, regio-, diastereo-, and enantioselectivities of chemical reactions occurring in the cage-confined spaces, and distinct reaction selectivity may be achieved owing to variations in cavity microenvironments,^{11–17} offering new opportunities for the diverse synthesis of target molecules to address practical synthetic challenges faced under traditional homogeneous solution conditions. Recently, photosensitive metal–organic cages (PMOCs) have emerged as a promising paradigm of photocatalysts for eco-friendly organic transformations, especially for those unfavored and unconventional photochemical transformations of common compounds by the aid of

cage-confined photocatalysis.^{18,19} However, the asymmetric organic transformations catalyzed by cPMOCs with general substrate scope remain limited, mostly due to the specificity of host–guest binding in the closed central cavity and the difficulty in late-stage modifications of the cage size and binding sites.

The $\text{Ru}(\text{ImPhen})_3$ -based PMOCs developed by our group could mimic enzymatic activity by integrating multiple functional centers to form multiple cage pockets that allow dynamic guest-binding and exchanging.²⁰ The chiral open pockets of cPMOCs are able to prearrange planar aromatic substrates alongside the photoactive Ru-centers, leading to enhanced performance in asymmetric photochemical reactions. Recently, we have accomplished an uncommon cascade photochemical transformation from tropolone and chalcone with catalyst Δ_8 -/ Λ_8 -MOC-16- Ru_8Pd_6 (abbr. Δ -/ Λ -cage 1) to form bicyclo[3.2.2]nonenes,²¹ demonstrating the efficiency and superiority of the cage-confined catalysis in achieving outstanding chemo-, regio-, and enantioselectivities. The

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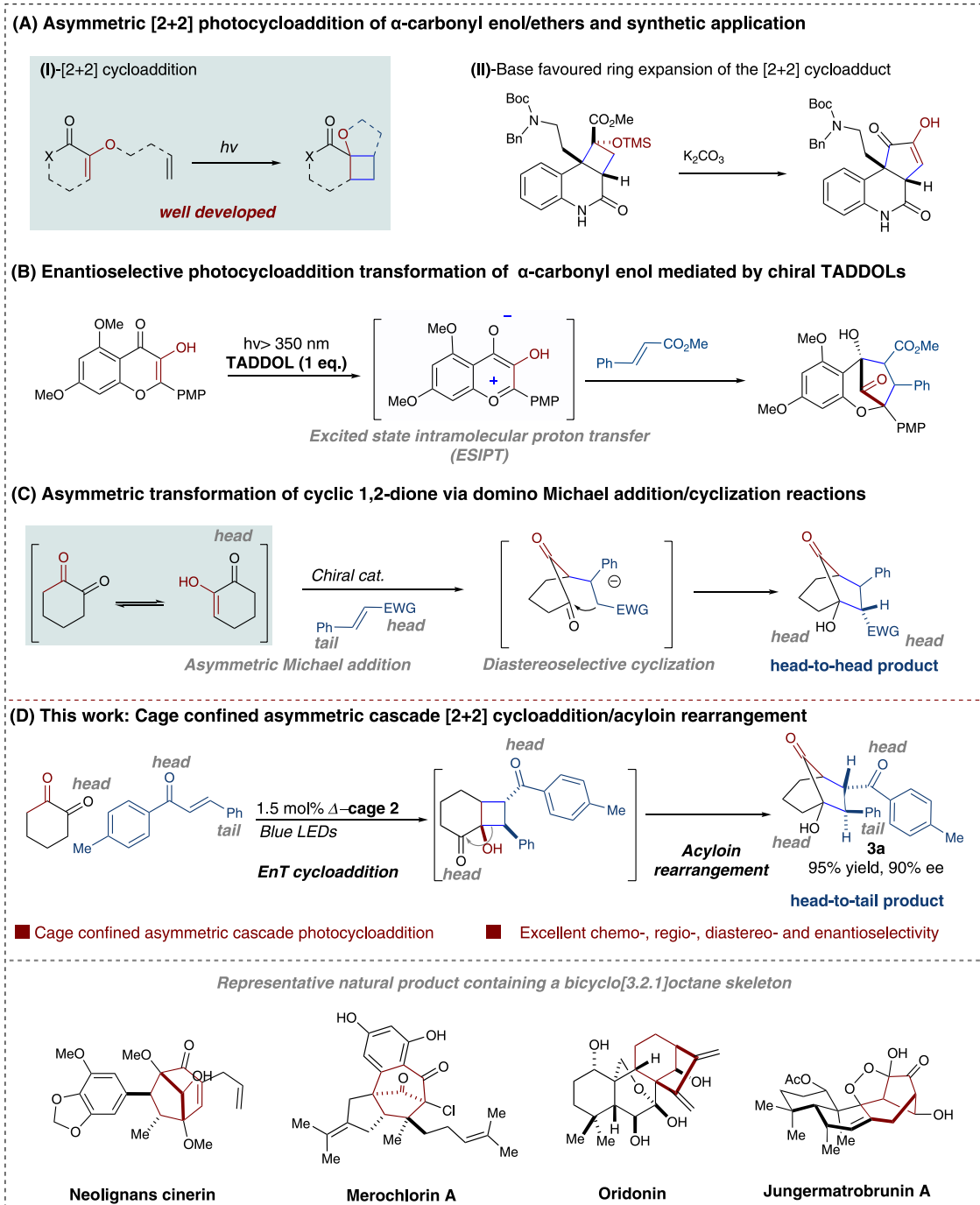


Figure 1. (A) Asymmetric [2 + 2] photocycloaddition of α -carbonyl enol/ethers and synthetic application. (B) Enantioselective photocycloaddition mediated by chiral TADDOLs. (C) Asymmetric transformation of cyclic 1,2-dione via domino Michael addition/cyclization reactions. (D) Cage-confined asymmetric cascade [2 + 2] photocycloaddition/acyloin rearrangement.

critical step involves adding a triplet-excited chalcone to the cyclic α -carbonyl enol or alkyl ether, followed by an acid-promoted 1,2-acyloin rearrangement and subsequent cascade transformations. However, attempts to isolate the 1,2-acyloin rearrangement product and broaden the substrates to more general cyclic α -carbonyl enols/ethers were unsuccessful, suggesting that host–guest matching interactions may predominate the reaction process.

Although the asymmetric [2 + 2] photocycloaddition transformation of the triplet-excited α -carbonyl enols/ethers has been accomplished by Bach,²² Yoon,²³ and Meggers et al.²⁴

through design of ingenious dual-functional chiral photocatalysts, the catalytic systems that can motivate the cascade asymmetric [2 + 2] cycloaddition/acyloin rearrangements remain rare, so well-designed substrates or additional treatments are usually needed to realize the ring expansion (Figure 1A(II)).^{25,26} Notably, the α -carbonyl enol could also be activated under photoirradiation, as illustrated by Porco et al., who used a stoichiometric amount of TADDOLs as a chiral source to mediate the asymmetric [3 + 2] cycloaddition transformation of 3-hydroxyflavone and cinnamyl ester,^{27,28} and proposed generation of the key oxidopyrylium inter-

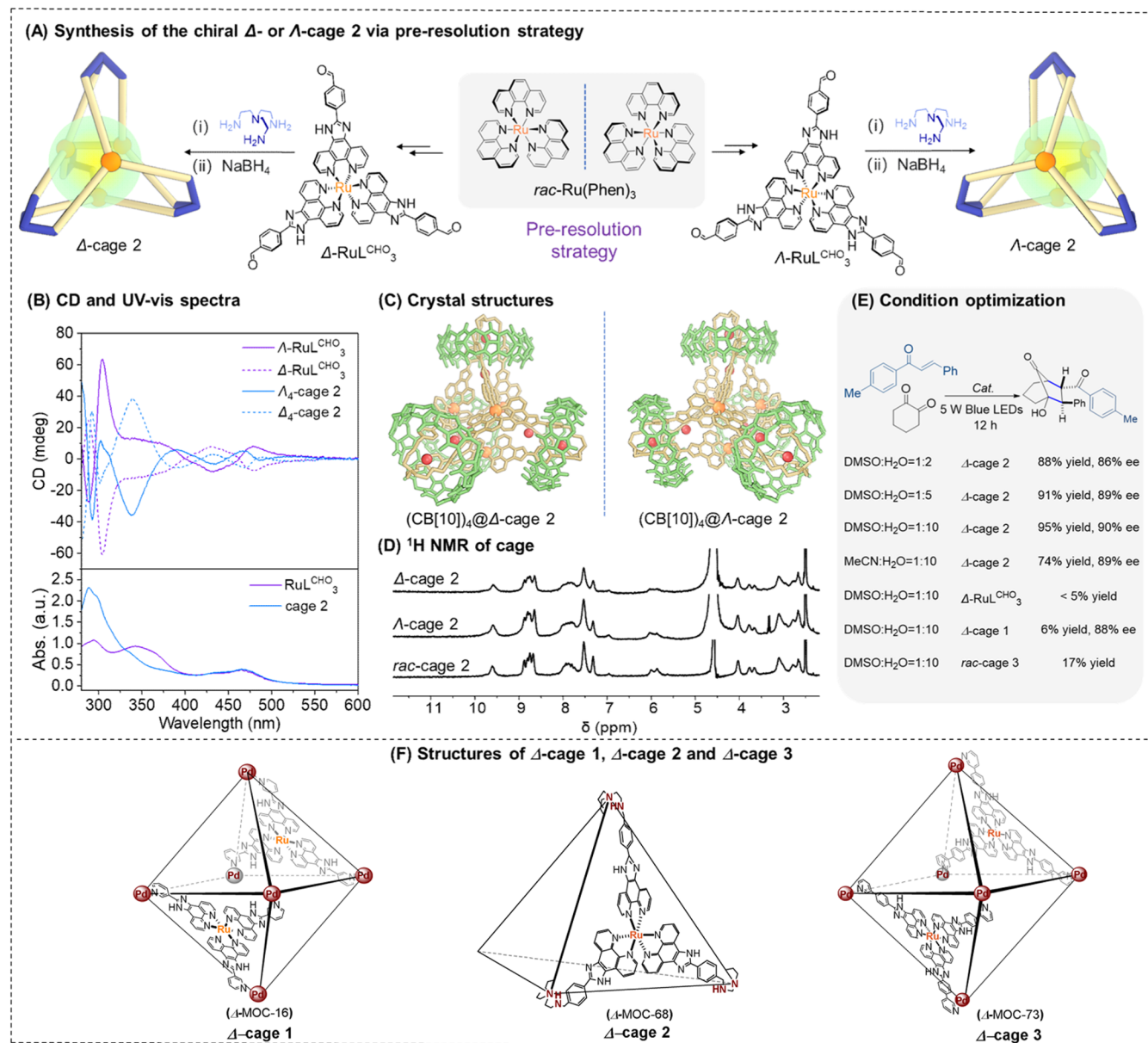


Figure 2. (A) Synthesis of the chiral Δ - or Λ -cage 2 via pre-resolution strategy. (B) Circular dichroism (CD) and UV-vis spectra of Λ -RuL₃CHO, Δ -RuL₃CHO (DMSO, 4×10^{-5} M), Λ -cage 2, and Δ -cage 2 (DMSO, 1×10^{-5} M). (C) Crystal structures of (CB[10])₄@ Δ -cage 2 and (CB[10])₄@ Λ -cage 2. (D) ^1H NMR comparison of Δ -cage 2, Λ -cage 2, and rac-cage 2 (DMSO-*d*₆/D₂O, 1/2, v/v). (E) Condition optimization. (F) Structures of Δ -cage 1, Δ -cage 2 and Δ -cage 3.

mediate through the excited-state intramolecular proton transfer (ESIPT) process (Figure 1B).²⁹ This means the catalytic asymmetric transformations of α -carbonyl enols/ethers involving visible-light-induced triplet-state donors would become even hard. Furthermore, the varied regioselective outcomes of the cascade photocycloaddition and acyloin rearrangement can lead to the formation of numerous unpredictable isomers,³⁰ posing challenges in achieving the terminal product with high chemo-, regio-, and diastereoselectivities. Despite cyclic 1,2-diones that could generate active α -carbonyl enol intermediate are able to act as “formal 1, 3-dipole” precursors to cycloaddition with the electron-deficient alkenes, and a series of efficient asymmetric cascade reactions based on Michael addition^{31,32} have been developed to construct bridged compounds under thermodynamic conditions^{33,34} (Figure 1C), the photocatalytic asymmetric

transformations of cyclic 1,2-diones remain exceedingly rare, and no report of visible-light-induced enantioselective photochemical transformations of cyclic 1,2-diones is known yet.

Because the highly positive-charged Ru(ImPhen)₃-based cages could in situ create outer-acidic and inner-basic amphoteric characteristics in solution,³⁵ which will facilitate both the base-driven conversion of cyclic 1,2-diones into α -carbonyl enols and the acid-driven 1,2-acyloin rearrangement, we envision the asymmetric photochemical transformation of cyclic 1,2-diones could be achieved by the cage-confined catalysis via photocycloaddition/acyloin rearrangement if the chiral cages possess adaptable pockets to match prerequisite host–guest interactions. Herein, we present our recent achievement in the visible-light-induced asymmetric cascade cycloaddition/acyloin rearrangement by developing the new cPOMC as photoreactors, generating chiral bicyclo[3.2.1]-

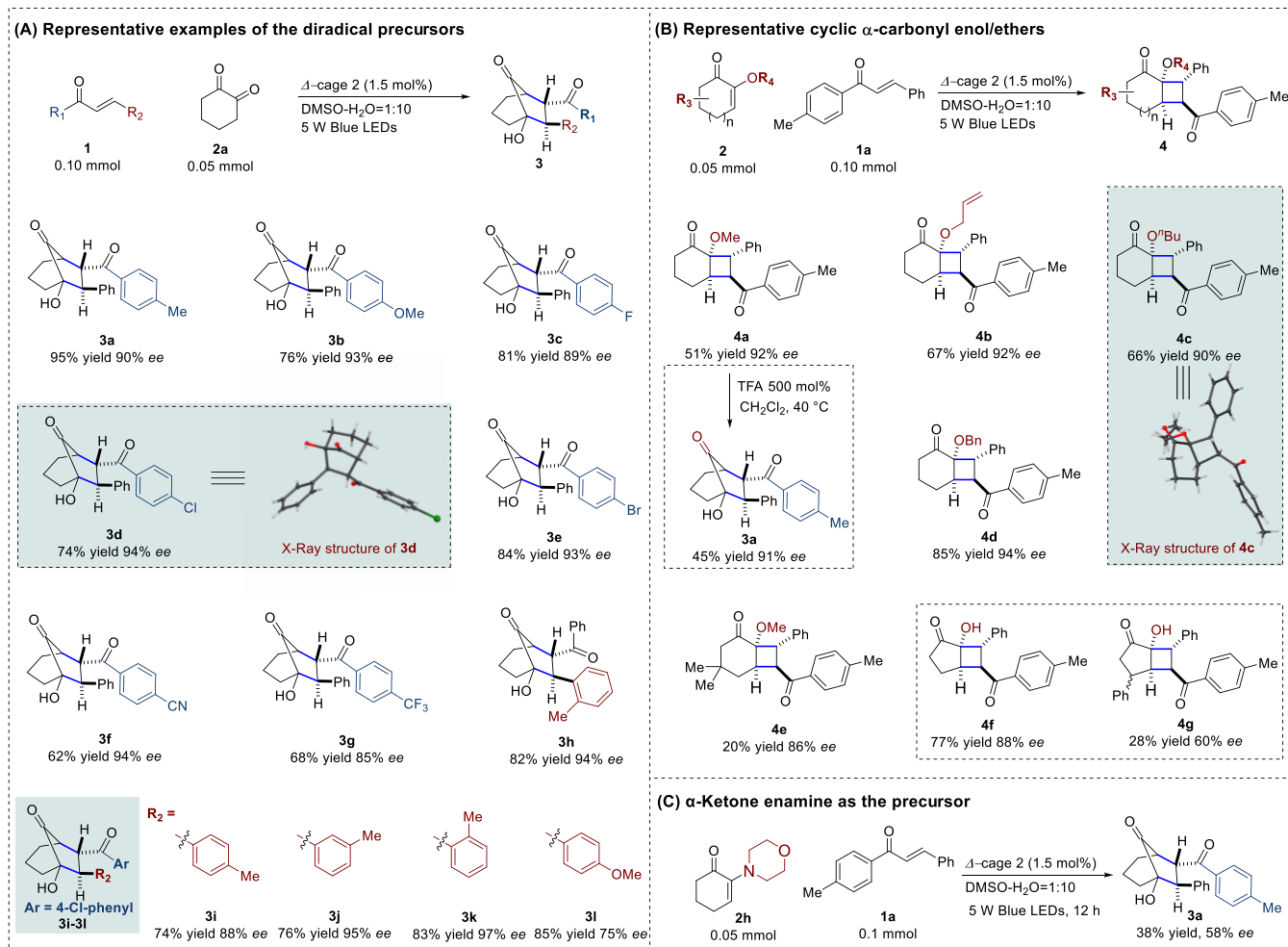


Figure 3. Substituent effect at different positions of substrates. (A) Representative examples of the diradical precursors. (B) Representative cyclic α -carbonyl enol/ethers. (C) Reaction with α -ketone enamine as the precursor.

octanes which are crucial subunits of many bioactive natural products and drugs (Figure 1D).^{36–42} A pre-resolution strategy was employed to obtain the enantiomers of chiral Ru-(ImPhen)₃-based cages, Δ_4 -/ Λ_4 -MOC-68-Ru₄ (abbr. Δ -/ Λ -cage 2), which differed from Δ_8 -/ Λ_8 -MOC-16-Ru₈Pd₆ in their pocket shape and size (Figure 2).^{21,43,44} The distinct substrate binding behaviors resulting from pocket microenvironmental difference caused cage 2 to retard [2 + 2] homodimerization of chalcone substrates while facilitating heterocycloaddition with α -hydroxy ketones; subsequently, 1,2-acyloin rearrangement delivered the bicyclo[3.2.1]octane skeletons. Noteworthily, the photocycloaddition regioselectivity under cage-confined conditions is opposite to the cycloaddition transformation of cyclic 1,2-diones under classic thermodynamic conditions.

RESULTS AND DISCUSSION

Synthesis of Chiral Cage 2. We previously assembled *rac*-cage 2 from *rac*-RuL₃CHO, which were composed of Δ -cage 2 and Λ -cage 2 enantiomeric cages in equal amounts to give a racemic mixture.⁴³ This indicates that the synthesis of enantiopure cages needs the first acquisition of the chiral metalloligands. Benefiting from the kinetic inertness of the Ru-centers, a *pre*-resolution strategy can be adapted to obtain homochiral cages (Figures 2 and S1–S8).⁴⁵ Initially, *rac*-Ru(Phen)₃ was resolved using chiral K₂[Sb₂{(±)-tartrate}₂].

3H₂O to give Δ -Ru(Phen)₃ and Λ -Ru(Phen)₃, which were further oxidized and subjected to Debus-Radziszewski imidazole synthesis to produce Δ -RuL₃CHO and Λ -RuL₃CHO metalloligands, respectively. Circular dichroism (CD) spectroscopy indicated that the chirality of the Ru-centers was retained after oxidation and imidazole reaction (Figure 2B). The chiral Δ - or Λ -metalloligands were subsequently condensed with tris(aminoethyl)amine (TAEA) to form tetrahedral intermediate cages,⁴³ which were reduced by NaBH₄ to form Δ - or Λ -cage 2 metallo-amine cages with strong resistance to acid and base (Figure S1). ¹H NMR characterization confirmed that the enantiomeric cages synthesized via *pre*-resolution strategy were consistent with their racemic counterparts (Figure 2D), demonstrating that this method achieved high chemical purity. CD spectra further revealed enantiopurity of the obtained chiral cages (Figure 2B). Furthermore, cage 2 can assemble with cucurbit[10]uril (CB[10]) to lead to (CB[10])₄@ Δ -cage 2 and (CB[10])₄@ Λ -cage 2 host-on-host ensembles (Figure 2C), which show pseudorotaxane structures with CB[10] rings capping on the four vertices of the cage to preclude close intercage interactions. Consequently, both racemic and chiral cages exhibit similar stacking patterns in the presence of CB[10], which has been unambiguously clarified by the single-crystal X-ray diffraction analysis (Figures S5, S6 and Table S1). The *rac*-cage 2 shows an equal probability of Δ -cage 2 and Λ -cage 2

occupying the same positions in the crystal lattice, leading to overlapped site occupancy disorder.⁴³ In contrast, the chiral cages preserve uniform chirality without any disorder, showing homochiral cages in the whole single crystal, which further confirms the enantiopurity of Δ - or Λ -cage 2.

Asymmetric Cascade Catalysis with α,β -Unsaturated Ketones and Cyclic α -Carbonyl Enols/Ethers. In previous studies, we revealed that the rhombic dodecahedron cage 1 could achieve striking catalytic performance in both homo- and heterocycloaddition of two coplanar acyclic olefins, aided by the *pre-arrangement* effect in the confined cage pockets.^{46,47} Similarly, cage 2, possessing analogous intrinsically active pocket-sites, also exhibited excellent [2 + 2] photocycloaddition activity for α,β -unsaturated ketones, showing superior diastereoselectivity compared to cage 1 owing to the unique dynamic adjustment ability of this allosteric cage.⁴⁴ Given the excellent reactivity of cage 1 and cage 2 in photocycloaddition reactions, we wonder if they were potential candidates for the selective transformation of cyclic α -hydroxy ketones which could be generated in situ from cyclohexyl 1,2-dione in the basic inner cage-space, leading to unprecedented transformation models under the enzymatic cage-confinement effect.

First, we employed cage 1 as a photocatalyst for the cycloaddition between α,β -unsaturated ketone **1a** and α -hydroxy ketone **2a**, the reaction predominantly produced a significant amount of homodimerization product (Figure S9 and Table S3, entry 8). This indicates that under the conditions of Δ -cage 1, the nonaromatic substrate **2a** hardly participated in the photocycloaddition reaction. By contrast, when switching to photocatalyst Δ -cage 2, the heterocycloaddition was significantly improved under optimized conditions, achieving yields up to 95% and excellent 90% enantiomeric excess (ee) for product **3a** (Figure 2E). This illustrates that, despite similar photosensitive Ru-centers, the differences in cage structures and pocket sizes can lead to notably different reactivity. The control experiment using the cage-precursor metalloligand Δ -RuL₃^{CHO} as catalyst resulted in only trace quantities of the desired product **3a**, which underscores the crucial role of the cage-confined cavities in enhancing reaction kinetics and governing selectivity. In the absence of light, the reaction proved infeasible, thereby confirming its dependence on photoexcitation (Table S3).

Substrate Scope. With the optimized conditions in hand, we then endeavored to investigate the substrate scope of chalcones (Figure 3A). This photocatalytic system demonstrated commendable functional group tolerance, accommodating both electron-withdrawing and electron-donating substituents. When the benzoyl group was substituted at the *para* position with an electron-donating group such as methoxy, the target product **3b** was obtained with 76% yield and 93% ee. Even upon substitution with electron-withdrawing groups such as halogens, the system maintained good yields and high enantioselectivity (**3c**–**3e**). The absolute configuration of these products was confirmed through X-ray single-crystal diffraction analysis of **3d** (Figure 3A). However, substituents exhibiting pronounced electron-withdrawing effect, such as trifluoromethyl and cyano groups, led to an increase in dimerization byproducts of chalcone, which slightly diminished the yields. Next, we examined the influence of substituents on the aromatic ring regarding reaction efficiency. Notably, when a methyl substituent was present at the *ortho* position, the product was obtained with an 82% yield and 94%

ee (**3h**), indicating that the steric effect induced by the torsion of the aromatic ring favors the generation of a single isomer. Substitutions at the *meta* and *para* positions with methyl group also gave favorable results (**3i**, **3j**). However, when a strong electron-donating group such as methoxy was located at the *para* position, the enantioselectivity decreased to 75% ee (**3l**). Conversely, the presence of strong electron-withdrawing groups on the aromatic ring hindered the ability to be sensitized due to the elevation of triplet-state energy, making it challenging to obtain the target product.

When the hydroxyl group of the cyclic 1,2-dione was protected as ether or ester, the rearrangement reaction did not proceed smoothly, resulting in products of a bicyclic[4.2.0] structure formed through intermolecular [2 + 2] photocycloaddition (Figure 3B).^{48–50} Substitutions with methyl (**4a**), allyl (**4b**), and *n*-butyl groups (**4c**) led to slightly reduced yields. In contrast, using a bulkier benzyl substituent (**4d**) resulted in suboptimal yield but remarkable enantioselectivity (94% ee), suggesting that the steric effect caused by the substituents on 1,2-diones also played an important role in photocycloaddition reactivity and selectivity. The X-ray single-crystal structure of product **4c** elucidated the absolute configuration of these compounds, indicating that the *cis*-migration of C–C bond is congruent with the configuration of the preceding rearrangement reaction, thereby confirming the involvement of a four-membered ring intermediate for the above cascade transformation. We also explored the introduction of substituents on the six-membered ring. Specifically, when position 5 was substituted with *gem*-dimethyl groups, the yield plummeted to poor 20% for **4e**. This decline was likely attributable to the steric effect of the *gem*-dimethyl group on the ring, which may obstruct the entry of the six-membered ring into the cage pocket. Building on these results, we sought to use a structurally similar five-membered ring, 1,2-cyclopentanedione, as a 1,2-dione precursor. Unlike the unsubstituted six-membered ring, when using **2f**⁵¹ as the substrate, no rearrangement occurred to give bicyclo[2.2.1]heptane. Instead, the structure was confirmed as a [2 + 2] product **4f** with a good 77% yield and 88% ee, which further demonstrated that cyclobutanol was involved as an intermediate in the above cascade photochemical transformations. When using the prochiral compound 4-phenylcyclopentanedione (**2g**) as a substrate, both yield and enantioselectivity significantly decreased because the bulky group on the cyclic alkyl ring hindered the interaction between the two substrates within the confined space. Noteworthy, the α -ketone enamine **2h** also effectively acted as an acceptor for the excited chalcone, producing **3a** as the final product which should follow hydrolysis of the imine intermediate (Figure 3C). To understand the driving force behind the acyloin rearrangement, we tried to remove the methyl group from the bicyclic compound **4a** under acidic conditions, and the rearranged product **3a** was obtained. This result was consistent with the preceding cascade photochemical transformations, proving that the cage-catalyzed bicyclo[3.2.1]octane products originated from the [2 + 2] cycloadducts via 1,2-acyloin rearrangement.

Mechanism Investigation. It is reasonable that the homodimerization of two α,β -unsaturated ketones and the heterocycloaddition between α,β -unsaturated ketone and 1,2-dione are a competitive process subject to the influence of the host–guest recognition behaviors of cage catalysts. Cage 1 shows significant preference for the homocycloaddition of **1a**

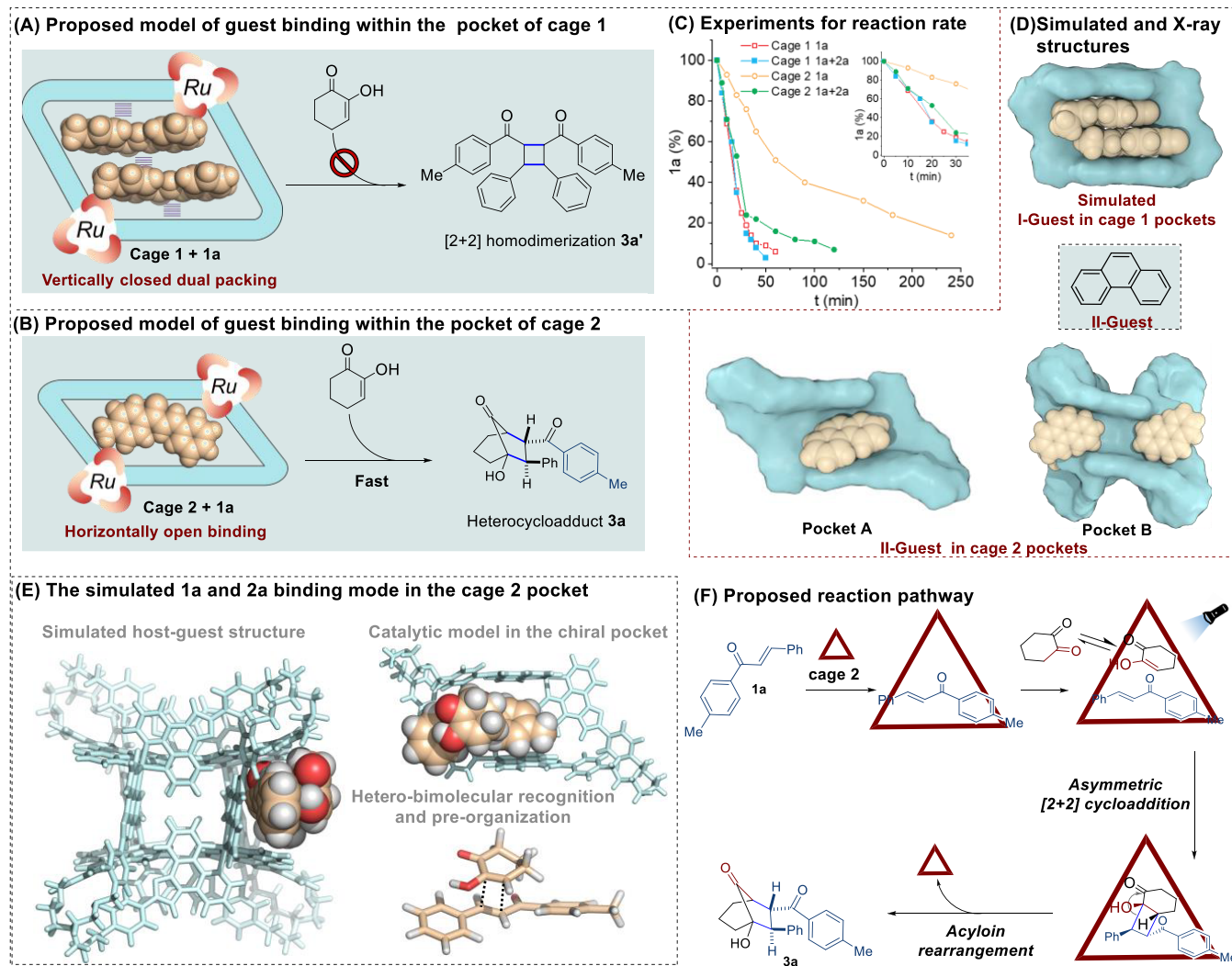


Figure 4. (A) Illustration of two **1a** guest-binding pattern in cage 1 pocket and the fast reaction of homocycloaddition. (B) Illustration of the horizontally open binding of one **1a** guest in the cage 2 pocket. The homocycloaddition of two **1a** is much slower than the heterocycloaddition of **1a** and **2a**. (C) Control experiment for the reaction rate determination. (D) X-ray single-crystal structures and simulated structures of the host–guest complexes. (E) Simulated **1a** and **2a** binding mode in the cage 2 pocket. (F) Proposed reaction pathway.

substrates^{46,47} in comparison with α -hydroxy ketones, whereas cage 2 evidently favors heterocycloaddition between substrates **1a** and **2a**. The different catalytic reactivity of cage 1 and cage 2 should primarily relate to their distinct cage structures,²¹ which will dominate their guest-binding behaviors due to their different cage pocket shapes and sizes (Figure S8). To gain direct vision of the guest-binding characteristics of cage 2, we obtained a phenanthrene $\subset$ *rac*-cage 2 single crystal for the host–guest structure understanding (Figure S7). Due to the size limitation of the cage pockets (8.8 Å in width), two planar aromatic guests cannot form vertically dual-packing mode as cage 1 of bigger pocket size (10.6 Å in width) does (Figures 4A, 4D and S8),^{52,53} but adopt a vertically single-guest binding mode in the empty pocket or a horizontally binding mode at two sides of the pocket containing one anion in the middle (Figure S7). The vertically dual-packing mode in cage 1 prefers cobinding of two planar acyclic **1a** substrates but hinders further contact of the additional nonplanar cyclic substrate **2a**, which basically explains why the homodimerization of **1a** predominates over heterocycloaddition between substrates **1a** and **2a** when catalyzed by cage 1. However, the horizontal

single-guest binding of **1a** by cage 2 makes the substrates open to access of the secondary substrate, allowing smaller-sized cyclic **2a** to contact, thus enabling cross-photocycloaddition. To validate this hypothesis, cage 3 with an even larger pocket size (12.8 Å in width, Figure S8)²¹ than cage 1 was also tested for photocatalysis (Table S3). Although cage 3 has a similar rhombic dodecahedron structure to cage 1, its larger pockets do not match the dual-packing of two **1a** substrates as good as cage 1 does; meanwhile, it makes it feasible to allow co-encapsulation of one planar acyclic **1a** and one nonplanar cyclic **2a**, thus giving an improved yield for the heterocycloaddition compared to cage 1.

To further understand the kinetics of two kinds of cages for their different catalytic reactivities, we compared the [2 + 2] photocycloaddition rates of sole substrate **1a** under identical conditions (Figure 4). In the same concentration of photo-sensitive Ru-centers, cage 1 and cage 2 catalyze at initial rates of $r_{1-1a-30\text{ min}} = 4.86 \times 10^{-4} \text{ mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$ and $r_{2-1a-30\text{ min}} = 1.44 \times 10^{-4} \text{ mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$, respectively. Evidently, cage 1 showed a more than 3-fold higher rate for the homodimerization of **1a** compared to cage 2, which is attributed to the

avored dual-packing of two aromatic planar guests in the cage 1 pockets. When a nonaromatic guest **2a** was introduced, the reaction rate in cage 1 relative to **1a** consumption remained almost unchanged at a rate of $r_{1-1a-2a-30\text{ min}} = 5.10 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$, suggesting that the second substrate **2a** struggles to participate in the cage catalytic process of cage 1 but could not compete with the dimerization of chalcones (Figure 4A,4C). However, in the case of cage 2, the introduction of the secondary substrate **2a** noticeably increased the **1a** consumption rate more than three times to $r_{2-1a-2a-30\text{ min}} = 4.56 \times 10^{-4} \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$, indicating that the presence of **2a** significantly facilitates conversion of **1a** in a heterocycloaddition manner (Figure 4B,4C).

^1H NMR titration experiments in the mixture of DMSO- d_6 /D $_2$ O = 1:5 (v/v) were also conducted to examine the host–guest interactions between cage 2 and the substrates (Figure S10). The NMR titration spectra displayed significant changes in the proton signals of the aromatic rings as **1a** was gradually added, denoting that the binding site of **1a** is located at the aromatic pockets. By fitting the data using the Hill equation, we obtained a binding constant $K_a = 324 \text{ M}^{-1}$, demonstrating apparent binding affinity between **1a** and the cage. In contrast, the addition of **2a** resulted in negligible shifts of the proton signals of the host–guest system, suggesting much poorer binding behaviors due to the faster exchange of this small nonaromatic and nonplanar guest without the aid of π - π stacking interactions. Consequently, the **1a** substrates are likely to preferentially occupy the cage pocket over the small **2a** substrates in the mixed system, in agreement with the hypothesis that the photoreactions are dominated by prerequisite host–guest interactions between cages and substrate **1a**.

Due to the fast conversion of the substrate and the rapid guest exchange habit, it is challenging to obtain single crystals of cage 2 binding both substrate molecules. To elucidate the possible binding mode of two substrates in the cage pockets, we conducted computational simulations for the host–guest system (Figure 4E). The global energy minimization revealed that **1a** tends to bind horizontally in the single-guest mode within the cage 2 pockets through π - π stacking interactions, similar to the observed binding mode in phenanthrene C *rac*-cage 2 complex. The small-sized pockets of cage 2 make the bound **1a** more accessible to the smaller **2a** rather than another bigger **1a**, thus facilitating the hetero-photocycloaddition reaction via a hetero-bimolecular recognition process owing to complementary interactions between two differentiated guest molecules.^{16,47,54} Therefore, the proposed reaction mechanism for this cage-catalyzed cascade transformation is illustrated in Figure 4F. The first step starts with preferential binding of one substrate **1a** in a cage pocket, and then the secondary substrate **2a** enters the basic pocket space to be converted into α -carbonyl enol. Upon excitation of the cage by visible light, the triplet state of substrate **1a** is induced,^{46,47} which undergoes an enantioselective [2 + 2] photocycloaddition with enol-form of **2a** in the confined chiral pocket, delivering bicyclo[4.2.0]octane compounds as the intermediates/products. Due to the lack of planarity, these compounds would be expelled from the cage pocket via fast guest exchange, facilitating the turnover of the cage catalyst. Attributed to the outer-acidic condition of the cage solution, the produced cyclobutanol compound would further undergo acyloin rearrangement, thereby achieving the cascade asymmetric construction of the final bicyclo[3.2.1]octane skeleton.

CONCLUSIONS

In summary, we present the synthesis of a pair of enantiomeric Δ_4 -/ Λ_4 -MOC-68-Ru $_4$ cages (Δ -/ Λ -cage 2), which possess chiral photosensitive pockets. These chiral pockets manifest unique substrate binding behaviors to achieve visible-light-induced asymmetric [2 + 2] heterocycloaddition between α,β -unsaturated ketones and α -hydroxy ketones, affording diverse bicyclo[3.2.1]octanes through a cascade [2 + 2] photocycloaddition/acyloin rearrangement process. By protecting the hydroxyl group, the photoreaction could be terminated at the first cycloaddition step, affording the bicyclo[4.2.0]octanes as the products. This study not only achieved the first visible-light-induced asymmetric cascade [2 + 2] cycloaddition/acyloin rearrangement transformation but also provided a useful alternative photochemical method different from the thermodynamic methods for the advanced construction of chiral bicyclo[3.2.1]octanes. Additionally, it establishes a foundation for achieving enantioselectivity of complicated organics via cascade reactions by virtue of biomimetic supramolecular catalysis involving confinement effect and hetero-bimolecular recognition. Further exploration of new cage catalysts and synthetic application of current cage-confined photocatalysis will be disclosed in the future.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.Sc03074>.

Additional experimental details, methods, and crystallographic data, including NMR spectra for all compounds (PDF)

Accession Codes

Deposition Numbers 2372338–2372339 and 2390659–2390661 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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